Showing Changes Over Time Transcript

Welcome back. Today we will be discussing about taxonomy of data visualisation methods and in this we will be discussing topics on comparison of different type of plots. In today's lesson we will be covering different types of plots and graphs and how to use them for specific cases.

As all of you know changes over time is a key element of business analysis. All of us make comparison not just at one point in time or a period of time. Growth is essence of life therefore seeing the changes over a period of time is a very natural thing to understand phenomenon.

In this regard the best charts that you have are line charts. Line charts represent continuous variables and time is considered a continuous variable on which you can compare the growth of different aspects or different categories of variables. So line charts are often used to show the continuous change over a period of time.

Similarly you can also use area charts which are also line charts in some sense but the portion between the line and the x-axis is generally shaded to indicate the entire region and we can use these two types of charts to indicate changes over time. And in line chart you can use multiple series to study the variables to a group of variables how they have performed over a period of time. It can be sales from different product units in your system in your business and in area chart it can be population growth of different countries or different states in India.

So these are few examples of variables which are studied over a period of time and represented using line and area charts. Now like in the last lecture we will be seeing some special charts which can be used for this purposes so that you can build custom visualisations. We will now study stacked area chart.

It is similar to stacked bar chart but in this case the distinction is the variable is continuous and it is like area charts stacked on one another. We will now study stacked area chart. Stacked area chart is similar to stacked bar chart.

It will reveal the composition within a category and also allow us to change study the changes over a period of time. As you can see on the graphic on the right of the slide it is a stacked area chart representing the viewership on television, newspaper, radio and internet. This clearly shows over a period of time the television viewership has fallen replaced by internet while radio and newspaper have remained more or less constant with slight reduction.

You can understand the change in composition over a period of time and clearly bringing the distinction between reduction in television ratio versus increase in internet. All these details are captured in a single visualisation using stacked area chart and also it gives the it also tries them. It also builds on the gestalt principle of continuity wherein the continuous area represents a trend which is easily pursued by the human eye.

You can also use stacked area chart for representing compositions and when you know and how the composition is changing over a period of time it's a very preferred chart. Similar to the stacked area chart you have a steam chart however instead of areas getting stacked on one another steam chart indicates the flow of a variable across a period of time. It is very essential to study such relationships which indicate what are the preferences towards different categories of variables.

As in this example you can see the historical energy mix of Great Britain where you have black coal in the earlier 1950 period which has now been taken over by oil, gas and nuclear energy and renewables are also starting to take some space in the energy mix. This kind of charts where the composition is not fixed it is dynamic and evolving is very useful for presenting the changes over a period of time. The categories have to remain constant if you are using a stacked chart and now you can always add new categories and see how they are evolving.

Here the interesting part is the width of the stream indicates the intensity of that particular category any given point of time. As the stream narrows the intensity or composition of that particular category in the overall scheme has reduced. This kind of details can be presented in this steam chart and these are used heavily for understanding hydrological data where you can see the flow data can be easily represented.

In this chart the functional purpose is highlighted through peaks and troughs. As I mentioned the peak indicates very high intensity and trough represent fall in intensity or composition. We can understand these charts very intuitively for presenting flow variables.

Now we will see another chart which is used to study behaviour over a period of time. It is the flow map and I would like you to take a closer look on the visual on this slide. It is a very celebrated visual often taught in data visualisation courses.

This is the great march of Napoleon's army and he has moved from France and waged a campaign on Russia in Moscow facing the Russian winter and finally witnessing a defeat. This map depicts the Napoleon army at the start of the campaign with the width of the area that I am highlighting now and gradually as they move towards Russia the army has reduced significantly and the black part which represents the return of the army and finally you can see only a minuscule portion has returned to France. This clearly represents the disastrous campaign that Napoleon has led on Russia leading in loss of human lives and and the graph also in detail captures the change in temperatures showing the exposures that the army has faced resulting in loss of men and material.

It is very great in detail and one has to appreciate this map was drawn or this diagram was given when there were no computer aided graphics or other machinery that could do calculations on this scale. The detail with which it captures and the presentation schema are worth applauding and this is what a flow map represents. It shows how a particular variable characteristic has changed over a period of time.

For example we can understand the same kind of migration patterns between country or increase in species density in a particular location using flow maps. These are very useful examples and however as I mentioned these are custom visuals and can be used for specific.

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